
Software Requirements Specification

for

MealMitra

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CONTENTS.....	II
REVISIONS.....	II
1 INTRODUCTION.....	3
1.1 DOCUMENT PURPOSE.....	3
1.2 DOCUMENT CONVENTIONS.....	3
1.3 INTENDED AUDIENCE AND DOCUMENT OVERVIEW.....	3
1.4 PRODUCT SCOPE.....	3
1.5 REFERENCES AND ACKNOWLEDGMENTS.....	4
2 OVERALL DESCRIPTION.....	4
2.1 PRODUCT PERSPECTIVE.....	4
2.2 PRODUCT FUNCTIONALITY.....	4
2.3 USER CLASSES AND CHARACTERISTIC.....	5
2.4 OPERATING ENVIRONMENT.....	6
2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS.....	6
2.6 USER DOCUMENTATION.....	6
2.7 ASSUMPTIONS AND DEPENDENCIES.....	6
3 SPECIFIC REQUIREMENTS.....	7
3.1 EXTERNAL INTERFACE REQUIREMENTS.....	7
3.2 FUNCTIONAL REQUIREMENTS.....	8
3.3 USE CASE MODEL.....	8
4 OTHER NON-FUNCTIONAL REQUIREMENTS.....	10
4.1 PERFORMANCE REQUIREMENTS.....	10
4.2 SAFETY AND SECURITY REQUIREMENTS.....	10
4.3 SOFTWARE QUALITY ATTRIBUTES.....	10
4.4 DATABASE REQUIREMENTS.....	11
5 OTHER REQUIREMENTS.....	11

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a clear and detailed description of the Food Waste Management and Donation System named **MealMitra**.

This document outlines the system's purpose, features, operating environment, and functional overview. It serves as a reference for developers, testers, instructors, and stakeholders to understand what the system is expected to do and how it should behave.

1.2 Scope

MealMitra is a MERN stack web application that enables verified restaurants to donate surplus food and verified NGOs to request and collect it. The system supports user registration and verification, donation listing and management, pickup request handling, donation history tracking, administrative monitoring, and automated notifications. The platform is intended for restaurants/event organizers, NGOs, and administrators.

1.3 Definitions

Term	Description
NGO	Non-Governmental Organization
Admin	Administrator responsible for verifying users and monitoring the platform
JWT	JSON Web Token, used for authentication
MERN	MongoDB, Express.js, React.js, Node.js
Donor	Restaurant or event organizer donating surplus food
Receiver	NGO requesting and collecting donated food

2. Overall Description

2.1 User Characteristics

Restaurant (Donor)

- Basic computer and internet knowledge.
- Registers and uploads food donations.
- Manages pickup requests.
- Views donation history.

NGO (Receiver)

- Basic computer and internet knowledge.
- Searches available food donations.
- Requests food pickups.
- Tracks completed pickups.

Administrator

- Good understanding of system management.
- Verifies registered users.
- Manages donations and reports.
- Monitors overall system activity.

3. Functional Requirements

R.1 : User Registration and Authentication

R.1.1 User Registration

- **Description:** The system shall allow restaurants and NGOs to register by providing their organization details and uploading the required verification documents. Newly registered users shall remain in a **Pending** state until approved by the administrator.
- **Input:** Organization details, contact information, email, password, verification documents.
- **Output:** Registration successful message and account status set to **Pending Verification**.

R.1.2 User Login

- **Description:** The system shall authenticate registered users using their email and password. Only verified users shall be allowed to access the system dashboard according to their assigned role.
- **Input:** Registered email and password.
- **Output:** Successful login with role-based dashboard or an error message if credentials are invalid or verification is pending.

R.2 : User Verification

R.2.1 Verify Registered Users

- **Description:** The administrator shall review uploaded documents of restaurants and NGOs before granting access. This ensures that only genuine organizations participate in food donation activities.
- **Input:** User registration details and uploaded verification documents.
- **Output:** User status updated to **Verified** or **Rejected**.

R.2.2 Manage Verification Requests

- **Description:** The administrator shall be able to view all pending verification requests and approve or reject them with appropriate remarks.
- **Input:** Pending verification requests.
- **Output:** Updated verification status and notification sent to the user.

R.3 : Food Donation Management

R.3.1 Create Food Donation

- **Description:** Verified restaurants shall be able to create a food donation listing by entering food details such as quantity, food type, expiry time, and pickup location.
- **Input:** Food name, quantity, category, description, expiry time, pickup address.
- **Output:** New donation listing with **Available** status.

R.3.2 Update Donation

- **Description:** Restaurants shall be allowed to edit food donations before they are accepted by an NGO.
- **Input:** Updated donation details.
- **Output:** Donation details updated successfully.

R.3.2 Remove Donation

- **Description:** Restaurants shall be allowed to delete food donations before they are accepted by an NGO.
- **Input:** Deletion donation request.
- **Output:** Donation removed successfully.

R.4 : Food Browsing and Pickup Requests

R.4.1 Browse Available Donations

- **Description:** Verified NGOs shall be able to browse all available food donations and filter them based on location, food type, quantity, and expiry time.
- **Input:** Search keywords and filter criteria.
- **Output:** List of matching available food donations.

R.4.2 Request Food Pickup

- **Description:** NGOs shall be able to send pickup requests for available food donations. The request will be forwarded to the respective restaurant for approval.
- **Input:** Selected donation and pickup request.
- **Output:** Pickup request submitted successfully.

R.5 : Pickup Management

R.5.1 Manage Pickup Requests

- **Description:** Restaurants shall review incoming pickup requests and either approve or reject them based on availability.
- **Input:** Pickup request details.
- **Output:** Request status updated as **Accepted** or **Rejected**.

R.5.2 Track Pickup Status

- **Description:** The system shall update the donation status throughout the donation lifecycle, allowing both restaurants and NGOs to monitor the progress.
- **Input:** Status update actions.
- **Output:** Updated donation status such as **Available**, **Requested**, **Accepted**, **Picked Up**, or **Completed**.

R.6 : Donation History

R.6.1 Display Donation Records

- **Description:** The system shall store a complete history of donations made by restaurants and pickups completed by NGOs for future reference.
- **Input:** Completed donation information.
- **Output:** Permanent donation history accessible to authorized users.

R.6.2 View Previous Activities

- **Description:** Users shall be able to view their previous donations, pickup requests, and completed transactions through their dashboard.
- **Input:** User request to view history.
- **Output:** List of historical donation and pickup records.

R.7 : Administration and Monitoring

R.7.1 Manage Users

- **Description:** The administrator shall manage all registered users, including approving, rejecting, suspending, or removing accounts when necessary.
- **Input:** User management actions.
- **Output:** Updated user account status.

R.7.2 Manage Food Donations

- **Description:** The administrator shall monitor all active donations and remove invalid, expired, or inappropriate listings to maintain system reliability.
- **Input:** Donation management actions.
- **Output:** Updated donation records.

R.7.3 Generate Reports

- **Description:** The administrator shall generate reports containing information such as total registered users, total donations, completed pickups, and overall system activity for monitoring and analysis.
- **Input:** Report generation request.
- **Output:** Statistical reports and graphical summaries.

R.8 : Notification Management

R.8.1 Notify Users

- **Description:** The system shall notify restaurants and NGOs about important events such as account verification, pickup requests, request approvals, rejections, and completed donations.
- **Input:** System events.
- **Output:** Notifications displayed to the appropriate users.

R.8.2 Expiry Notification

- **Description:** The system shall automatically notify restaurants when a food donation is nearing its expiry time or remove expired donations from the available list.
- **Input:** Donation expiry information.
- **Output:** Notifications displayed to the appropriate users

4. Non-Functional Requirements

4.1 Performance Requirements

The MealMitra system shall provide quick access to food donation services by allowing restaurants to post donations and NGOs to browse and request pickups with minimal delay. The system should efficiently handle multiple users accessing the platform simultaneously. Performance depends on the number of active users, food listings, and pickup requests.

4.2 Safety & Security Requirements

The system shall ensure that user information, food donation records, and pickup requests are securely stored and protected from unauthorized access. Only verified restaurants, NGOs, and administrators shall be allowed to access their respective functionalities through secure authentication and role-based access control. User passwords shall be encrypted, and the system shall be regularly maintained to ensure reliable operation.

4.3 Software Quality Attributes

The system shall be reliable, user-friendly, and easy to maintain. It shall provide a simple interface that enables restaurants to donate food and NGOs to request pickups efficiently. The application shall be scalable to support future enhancements such as location-based matching, real-time

notifications, and a mobile application while maintaining consistent performance and data accuracy.

5. Constraints

- The system must be developed using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Authentication must be implemented using JSON Web Tokens (JWT) with role-based access control.
- Only verified restaurants and NGOs may create donations or send pickup requests.
- Styling shall use Tailwind CSS or Bootstrap.
- API testing shall be performed using Postman.
- Source code shall be version-controlled using Git and hosted on GitHub.

6. Assumptions

- Users have basic familiarity with using web applications.
- Restaurants and NGOs provide accurate registration and verification information.
- Stable internet connectivity is available to all users.
- The system depends on a functioning email/notification service for alerts

7. Software Requirements

- Frontend: React.js
- Backend: Node.js, Express.js
- Database: MongoDB
- Authentication: JWT
- API Testing: Postman
- Styling: Tailwind CSS / Bootstrap
- Version Control: Git & GitHub

8. Hardware Requirements

- Server: Cloud-hosted application server capable of running Node.js
- Client Device: Any desktop, laptop, tablet, or smartphone with a modern web browser

- Minimum Client Specification: Dual-core processor, 2 GB RAM, stable internet connection

9. User Interface Requirements

- Registration and Login pages for Restaurants and NGOs
- Restaurant Dashboard – donation management and pickup request handling
- NGO Dashboard – browsing, filtering, and requesting donations
- Admin Dashboard – user verification, donation monitoring, and reports
- Notification panel visible across all dashboards

10. Future Enhancements

- GPS-based nearby donation matching
- Real-time notifications
- AI-based demand prediction
- Mobile application
- Route optimization for pickups
- Integration with government food programs.